Survival Analysis

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Sessions

Monday 16th

Survival Analysis Fundamentals

core concepts
(2 hour sesion)

Wednesday 18th

Applications and Projects

exploring real-world uses
(1 hour sesion)

Monday 23rd

Coding a Real App

hands-on coding session
(2 hour sesion)

Wednesday 25th

Seminar Project Organization

project planning
(1 hour sesion)

TBD

Seminar Project Presentations

Final presentations of student projects

Evaluation

2

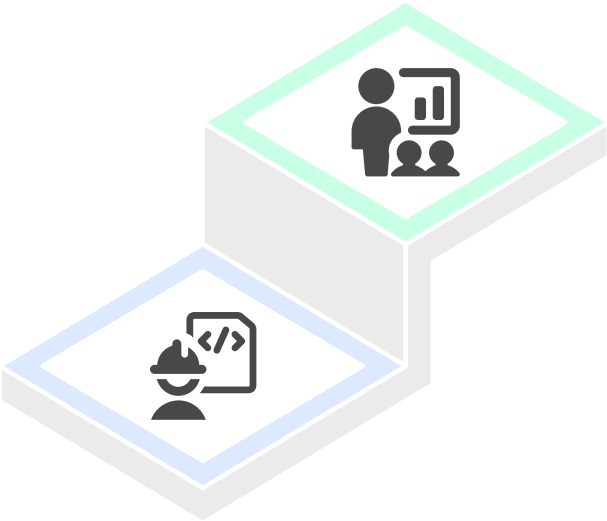
Seminar Project

Prepare a report and presentation.

1

Coding Activity

Refactor code and draw conclusions.



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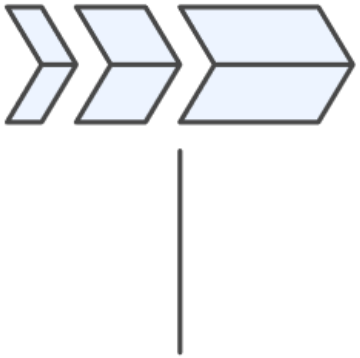

Survival
Analysis
Fundamentals

core concepts
(2 hour sesion)

What is Survival Analysis?

Context

Survival analysis in the real world
(time-to-event)

Pandemic disruptions to health care worsened cancer survival, study suggests

Mike Stobbe The Associated Press Feb 5, 2026 Updated Feb 5, 2026

FILE - A radiologist uses a magnifying glass to check mammograms for breast cancer in Los Angeles, May 6, 2010. (AP Photo/Damian Dovarganes, File)

DD MV**NY** PDS **NY** JN**NY**

Event: death/relapse (cancer survival)

[1] Canadian Press, Feb 2026. Source: https://www.thecanadianpressnews.ca/health/pandemic-disruptions-to-health-care-worsened-cancer-survival-study-suggests/article_06c13665-d37d-5cb5-a8c9-e5bbe4275dcb.html

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Event: default resolution
(time to resolution)

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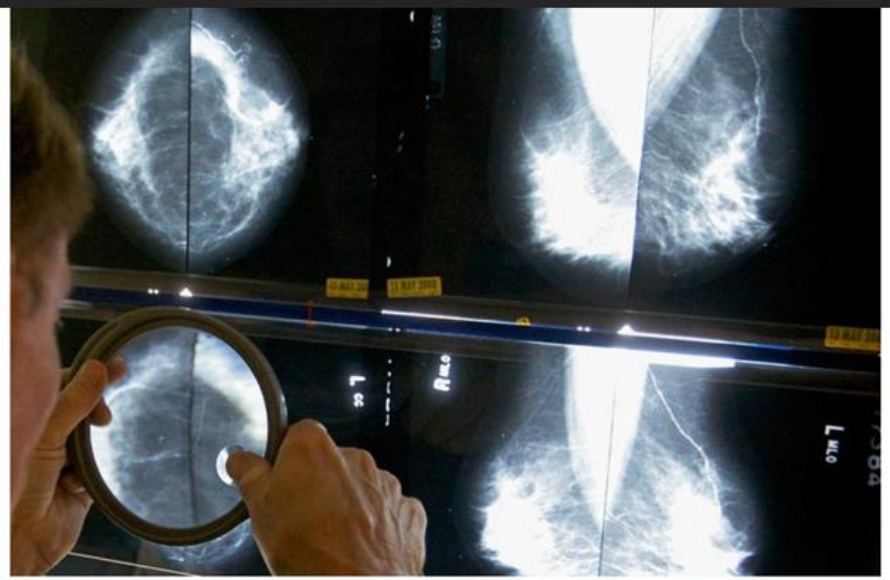

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NOVEMBER 25 2025

Rethinking default rates: Why time to resolution defines European credit opportunity

CONTRIBUTOR / GLOBAL NEWS, JOINT VENTURES AND PROMOTED CONTENT, , UK NEWS /

ARROW GLOBAL, ZACH LEWY

[1] Canadian Press, Feb 2026. Source: https://www.thecanadianpressnews.ca/health/pandemic-disruptions-to-health-care-worsened-cancer-survival-study-suggests/article_06c13665-d37d-5cb5-a8c9-e5bbe4275dcb.html
[2] Alternative Credit Investor, Nov 2025. Source: https://alternativecreditinvestor.com/2025/11/25/rethinking-default-rates-why-time-to-resolution-defines-european-credit-opportunity/

Context

Not just what event, but when!!

Survival analysis in the real world
(time-to-event)

Event: default resolution
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ARROW GLOBAL, ZACH LEWY

Shifting from preventive maintenance to predictive maintenance

ELE Times Research Desk January 15, 2026

Event: equipment failure (predictive maintenance)

[1] Canadian Press, Feb 2026. Source: https://www.thecanadianpressnews.ca/health/pandemic-disruptions-to-health-care-worsened-cancer-survival-study-suggests/article_06c13665-d37d-5cb5-a8c9-e5bbe4275dcb.html
[2] Alternative Credit Investor, Nov 2025. Source: https://alternativecreditinvestor.com/2025/11/25/rethinking-default-rates-why-time-to-resolution-defines-european-credit-opportunity/
[3] ELE Times, Jan 2026. Source: https://www.eletimes.ai/shifting-from-preventive-maintenance-to-predictive-maintenance

Index

The need for Survival Analysis

Is this a Regression Problem?

Fundamentals of Survival Analysis through a Use Case

How do we validate Survival Models?

Additional Use Case

What have we learnt?

What is Survival Analysis?

The need for Survival Analysis

- *Definition:* **Survival analysis** is a statistical method for analyzing **time-to-event** data, commonly used in healthcare for prognosis and risk estimation.

- *Example:* Suppose you are clinicians treating cancer patients and you need to decide which treatment is most appropriate for each patient.

The need for Survival Analysis

- If you were treating a cancer patient, you would not ignore individual characteristics...

- You would consider:
  - Age
  - Cancer stage
  - Biomarkers
  - General health condition
  - Type of treatment

- And you would ask: ***How long is this patient likely to survive?***
- Different patients → different risk profiles → different survival curves

The need for Survival Analysis

- Now imagine you are the CEO of a company hiring a new employee.

- You have data on:
  - Senior vs junior profiles
  - Years of experience
  - Industry background
  - Past job stability

- And you want to know: ***How long is this candidate likely to stay in the company?***
- Different profiles → different time-to-event distributions

(Event = resignation)

The need for Survival Analysis

- In all these cases, we observe an **event of interest** and our goal is to estimate **the time until it occurs**.

- *Examples of other real-world scenarios:*
  - Customer churn: time until a user cancels a subscription.
  - Equipment Failure: Time until a machine component breaks down.
  - Warranty Claims: Time until a product requires repair under warranty.

At First Glance: A Regression Problem?

- We would like to predict patient survival time using baseline characteristics and treatment type.

- This sounds like a standard supervised learning problem:

$$T = f(X) + \varepsilon$$

- Where:

$\boldsymbol{T}$**=*survival time***
$\boldsymbol{X}$**=*covariates***

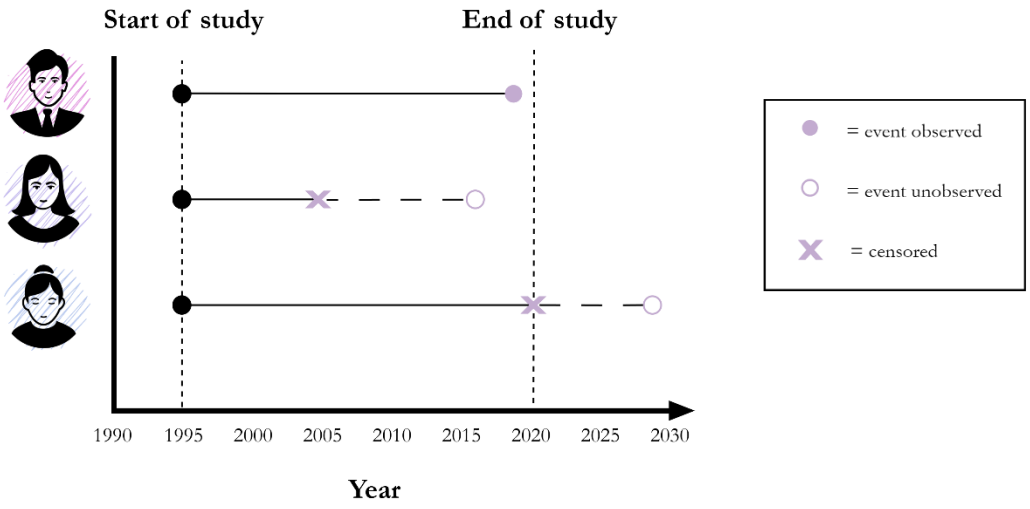
- Briefly remind what regression assumes:
  - The outcome variable is fully observed.
  - Each individual has a known response value.
  - Errors are well-defined for all observations.

At First Glance: A Regression Problem?

- However, in a five-year study, many patients may still be alive at the end of follow-up.

- For those patients:
  - We do **not** observe the true survival time.
  - We only know that their survival time exceeds the study duration.

PROBLEM:
Censored data



At First Glance: A Regression Problem?

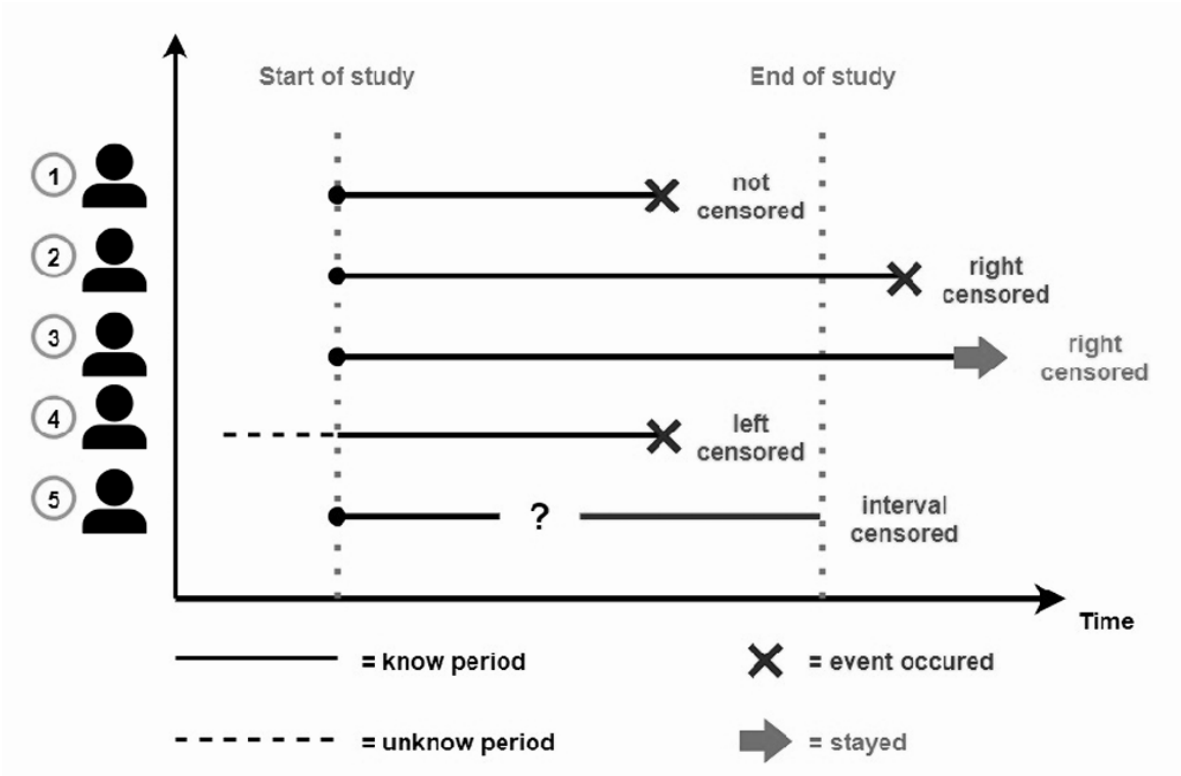
- If we applied in those cases ordinary regression:
  - We would treat censored times as exact values (incorrect).
  - Or we would remove censored individuals (we would bias the study, because we lose the 'survivors').
  - Or we would underestimate survival times.

- **Standard regression is not appropriate!!** ☹
  - We require a statistical framework that explicitly models time-to-event data with censoring → We need **Survival Analysis**

Survival Analysis Fundamentals

Censored Data

- Censoring: Event of interest not observed for all individuals.
- There are different types:



How Can Survival Analysis Handle Censoring?

- Censored Observations Are Not Ignored
  - Censored observations provide partial information
  - We know the individual survived **at least up to a certain time**

- In Survival Analysis, we look at a pair of variables for every individual:

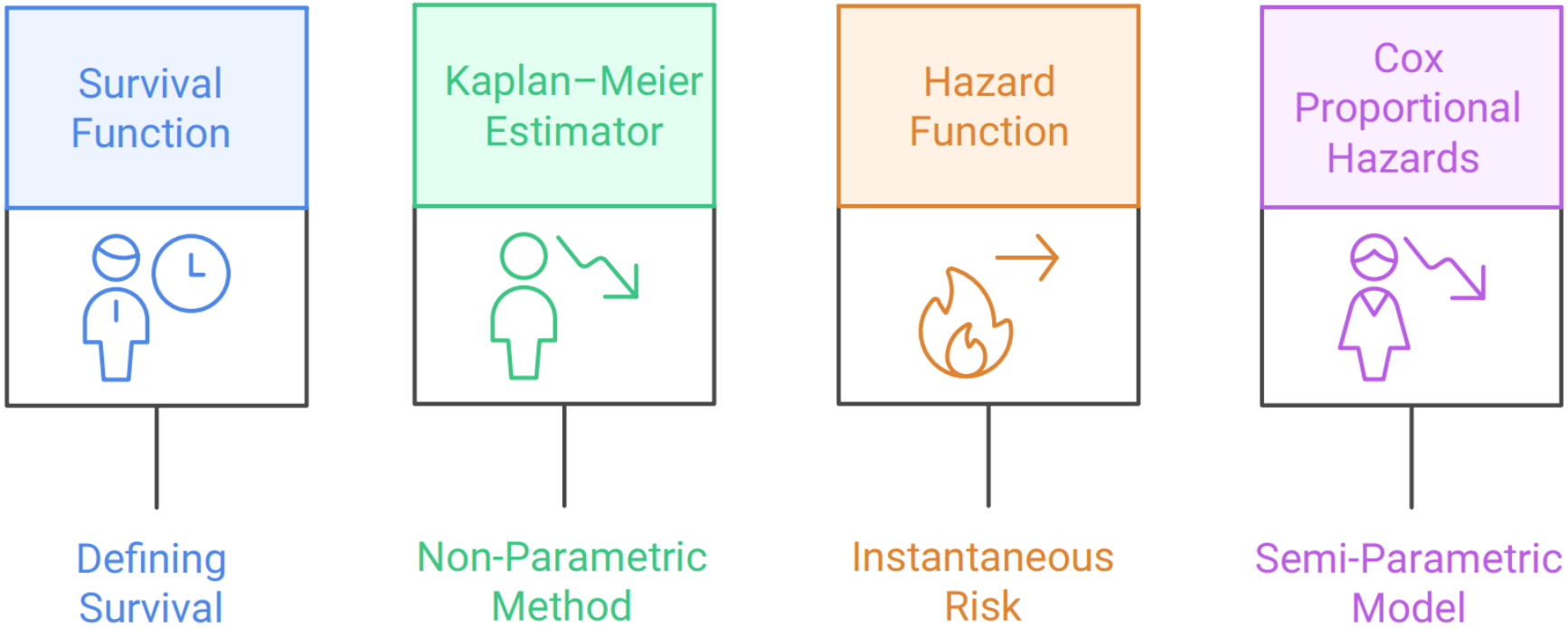
$$Y = \min(T, C)$$

- **Y (Observed Time):** This is the time we actually recorded. It is the minimum of the true **survival time (T)** or the **censoring time (C)**.

- **δ (Status Indicator):** A binary flag.
  - δ = 1: The event happened (uncensored).
  - δ = 0: The patient was censored (study ended, or lost to follow-up)

$$\delta = \begin{cases} 1, & if\ T \leq C \\ 0, & if\ T > C \end{cases}$$

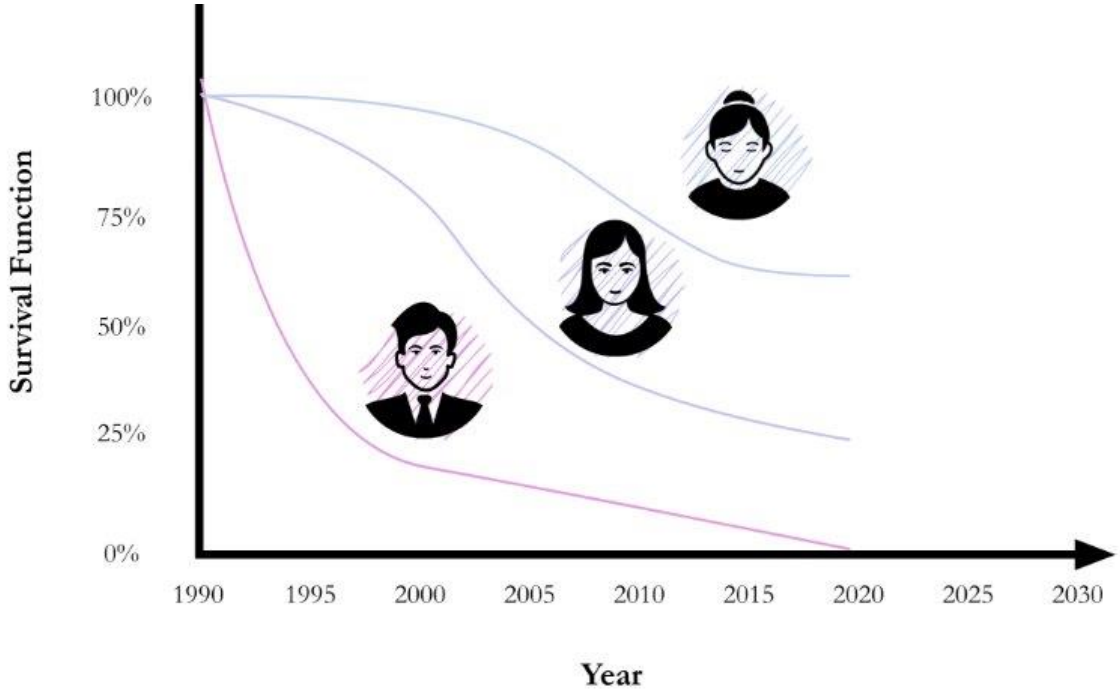
Core Concepts of Survival Analysis



Survival Function

The survival function returns the probability that the event has not occurred by time t:

$$S(t) = P(T > t)$$



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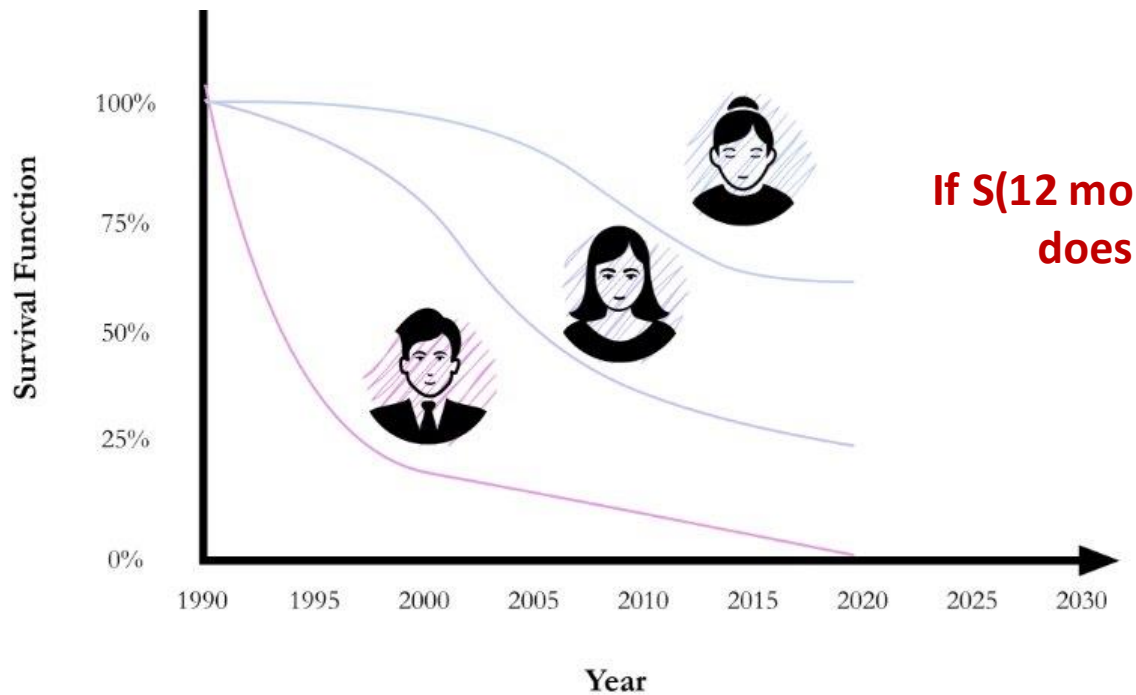

If S(12 months) = 0.8, what does this mean?

Kaplan-Meier Estimator

- How do we estimate the probability of surviving past time t?
- The Kaplan–Meier estimator[1] (1958) is a non-parametric method used to approximate the probability of survival over time.

Sir. Edward Kaplan

Sir. Paul Meier

[1] Kaplan, E. L.; Meier, P. (1958). "Nonparametric estimation from incomplete observations". J. Amer. Statist. Assoc. 53 (282): 457–481. doi:10.2307/2281868

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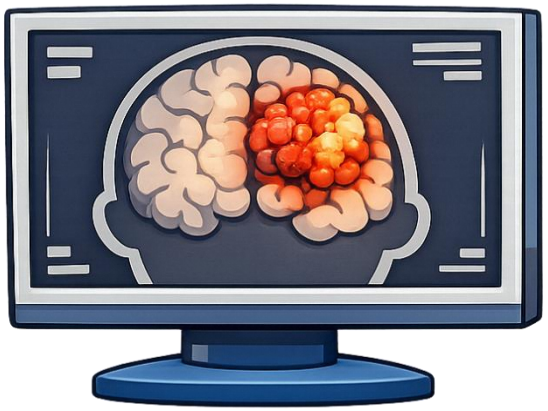
Why is it useful that the KM estimator is non-parametric?

[1] Kaplan, E. L.; Meier, P. (1958). "Nonparametric estimation from incomplete observations". J. Amer. Statist. Assoc. 53 (282): 457–481. doi:10.2307/2281868

Use Case 1: Brain Cancer

- The Brain Cancer dataset contains the survival times for patients with primary brain tumors.



Brain Cancer Analysis

GitHub

[1] Selingerová et al. (2016) Survival of patients with primary brain tumors: Comparison of two statistical approaches. PLoS One, 11(2):e0148733.

Use Case 1: Brain Cancer

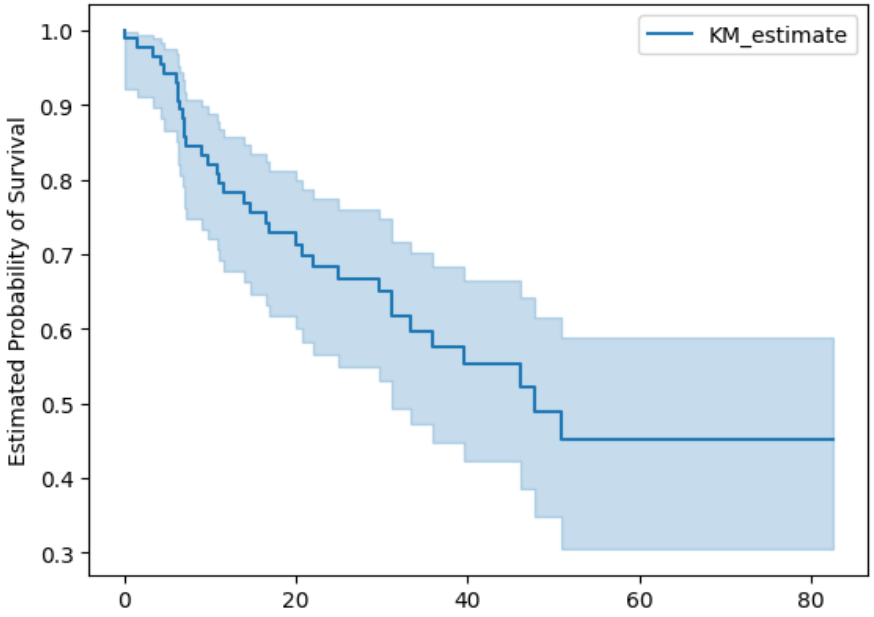
- Covariates
  - *gtv:* gross tumor volume, in cubic centimeters
  - *sex:* male or female
  - *diagnosis:* meningioma, LG glioma, HG glioma, or other
  - *loc:* the tumor location: either infratentorial or supratentorial
  - *ki:* Karnofsky index
  - *stereo:* stereotactic method: either stereotactic radiosurgery or fractionated stereotactic radiotherapy, abbreviated as SRS and SRT, respectively

	sex	diagnosis	loc	ki	gtv	stereo	status	time
1	Female	Meningioma	Infratentorial	90	6.11	SRS	0	57.64
2	Male	HG glioma	Supratentorial	90	19.35	SRT	1	8.98
3	Female	Meningioma	Infratentorial	70	7.95	SRS	0	26.46
4	Female	LG glioma	Supratentorial	80	7.61	SRT	1	47.80
5	Male	HG glioma	Supratentorial	90	5.06	SRT	1	6.30

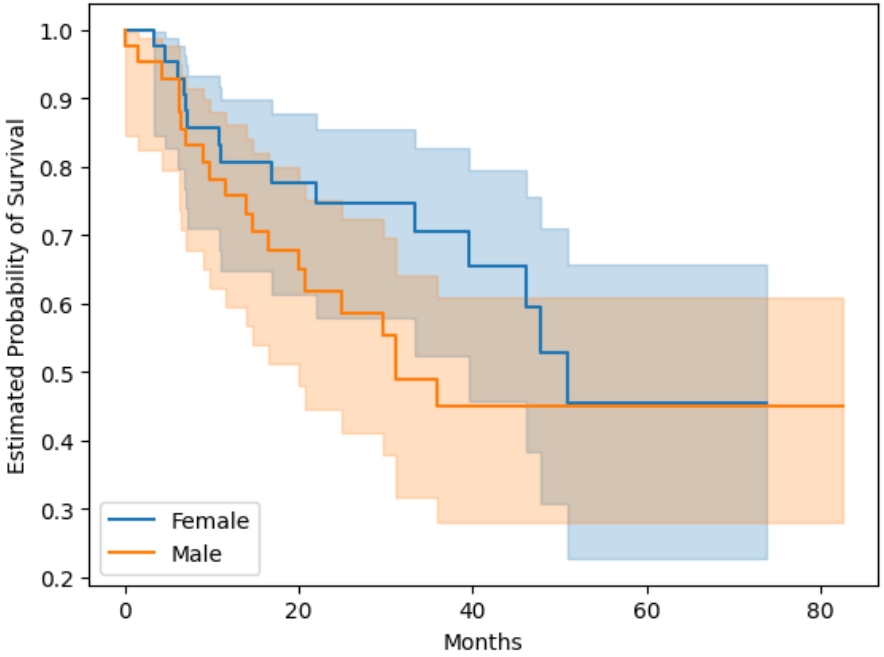
The Kaplan-Meier Curve

Brain Cancer Use Case

Kaplan-Meier survival curves



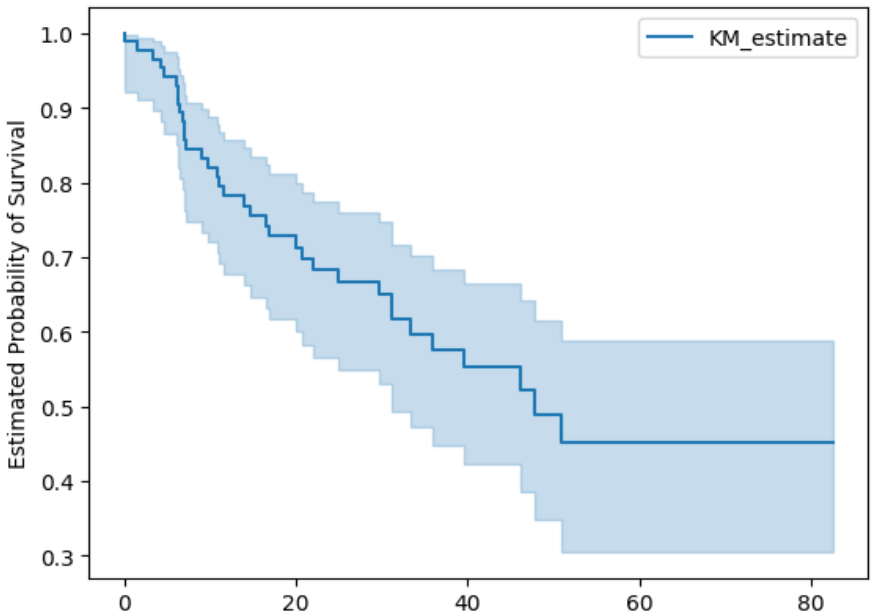
Kaplan-Meier survival curves stratified by sex



The Kaplan-Meier Curve

Brain Cancer Use Case

Kaplan-Meier survival curves



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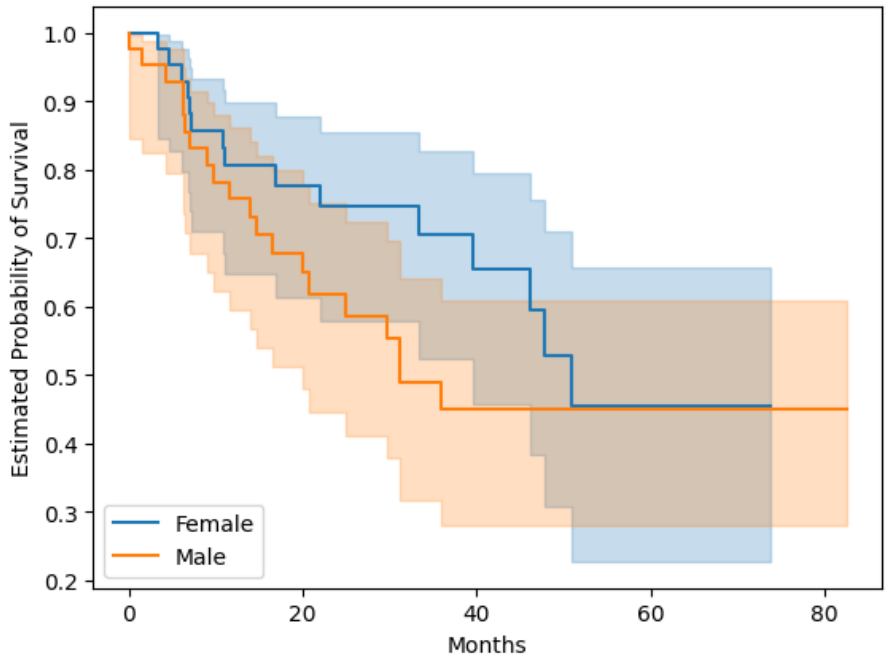


Curves look different, BUT we still need a formal test!!

Log-Rank Test

How can we carry out a formal test of equality of the two survival curves?

- *Why not use a t-test?*
  - Mean survival time cannot be directly compared using standard methods.
  - Assumptions of a two-sample t-test are violated.
  - Survival data include **censoring**.

- *Solution*: **Log-Rank Test**
  - Compares survival distributions between two groups.
  - Accounts for **censored observations**.
  - Evaluates whether observed events differ from expected events over time.

H_0: Survival curves are equal. If $p < 0.05$ → significant difference in survival

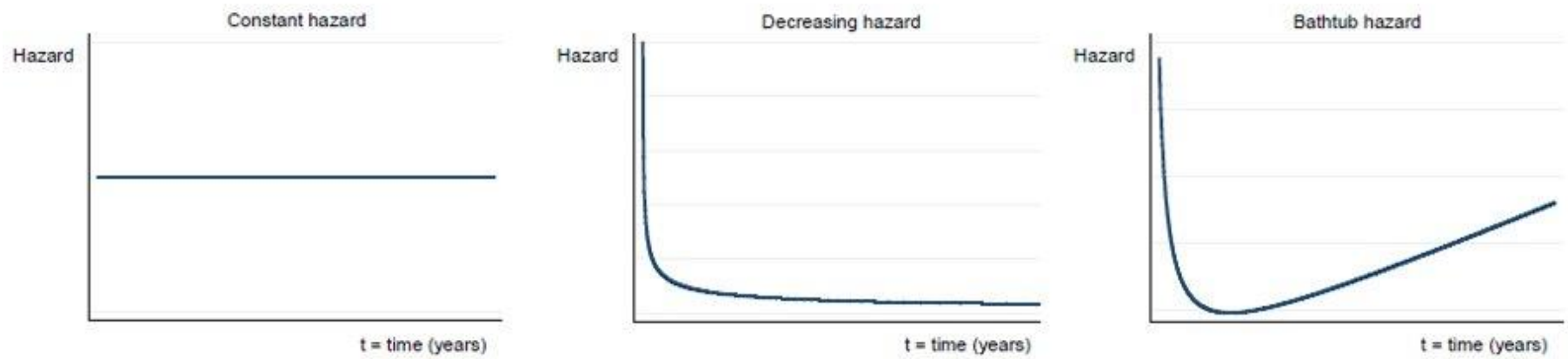
H_1: Survival curves differ. If $p \geq 0.05$ → no evidence of difference

Brain Cancer Analysis

GitHub

Hazard Function

The hazard function represents the instantaneous risk of the event occurring at time t:



Typical in engineering!

- *Why do we care about the hazard function?*

It turns out that a **key approach for modeling survival data as a function of covariates** relies heavily on the hazard function.

How is the Hazard Function Related to the Survival Function?

The hazard function represents the instantaneous risk of the event occurring at time t:

$$h(t|x) = \frac{p(t|x)}{S(t|x)}$$

where x are the covariates

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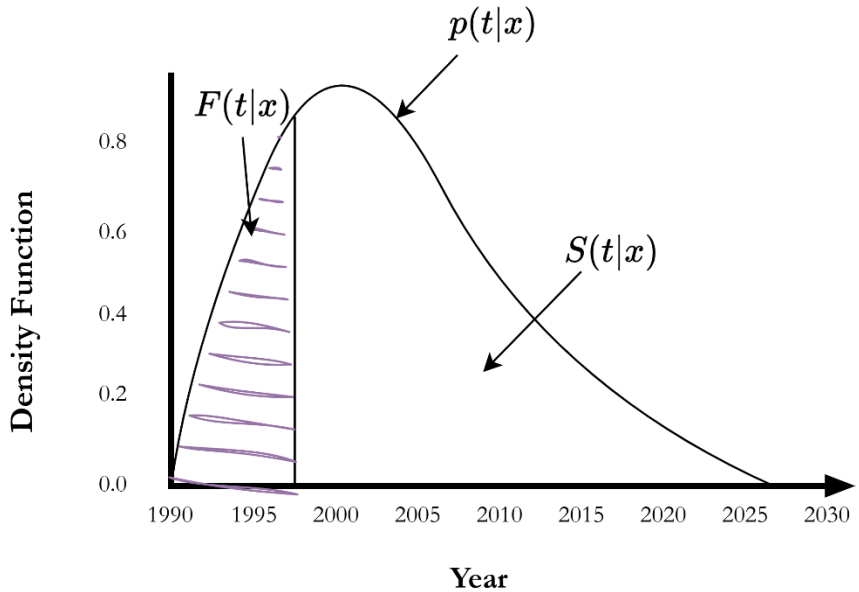
REMEMBER → *The Probability Density Function (pdf) $p(t|x)$ is the derivative of the Cumulative Density Function (CDF) $F(t|x)$. It can be rewritten as:*

$$p(t|x) = \frac{dF(t|x)}{dt} = S(t|x) \cdot h(t|x)$$

$$S(t|x) = P(T > t|x) = 1 - F(t|x)$$

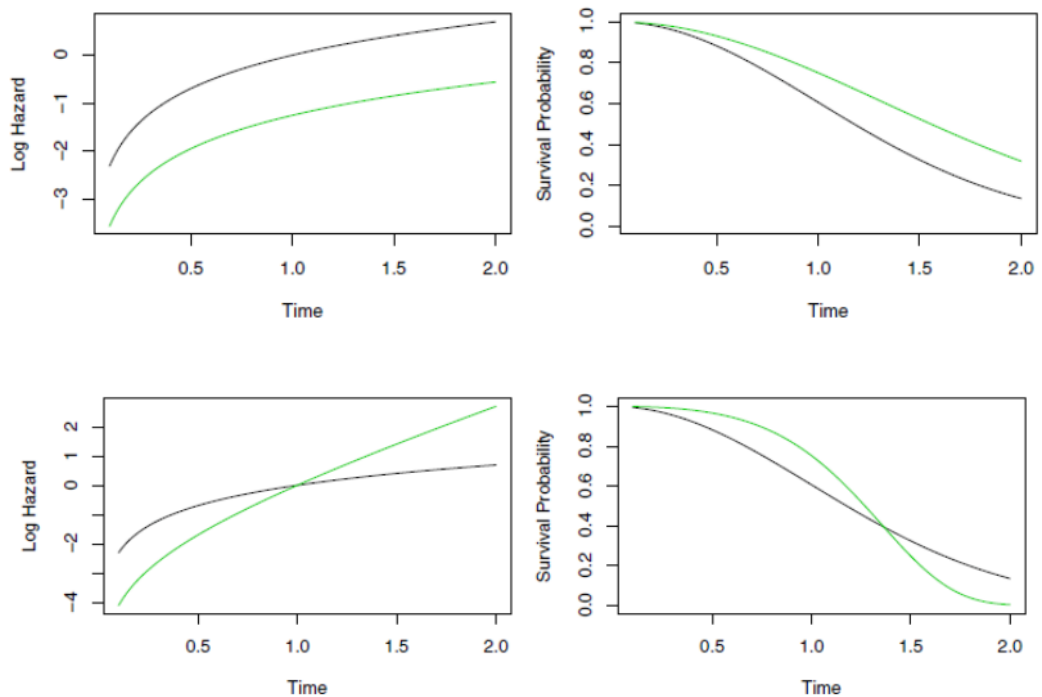


Proportional Hazards Assumption

- The *proportional hazards assumption* states: $$h(t|x_i) = h_0(t)\exp\left(\sum_{j=1}^{p} x_{ij}\beta_k\right)$$

- *Assumes* separate time dependence (baseline hazard h_0) from feature dependence

PH
Vs.
Non-PH



?

Can you think of a real situation in which PH is unrealistic?

If one group is riskier than another, it stays riskier by roughly the same factor throughout time.

Cox Proportional Hazards Model

- The **Cox Proportional Hazards**[1] model is a semi-parametric method used to assess the effect of one or more predictors on survival.

- *Unlike Kaplan–Meier*, it can **adjust for covariates** and estimate the relative risk (hazard ratio) associated with each predictor.

- The Hazard Ratio represents the relative risk of the event (example: death) occurring:
  - HR = 1: Neutral. The factor has no effect on the risk of death.
  - HR > 1: Increased Risk. The factor is associated with a higher risk (shorter survival).
  - HR < 1: Protective Effect. The factor reduces the risk (longer survival).

- **Remember!** Positive coefficient = Higher Risk = Lower Survival.

Brain Cancer Analysis

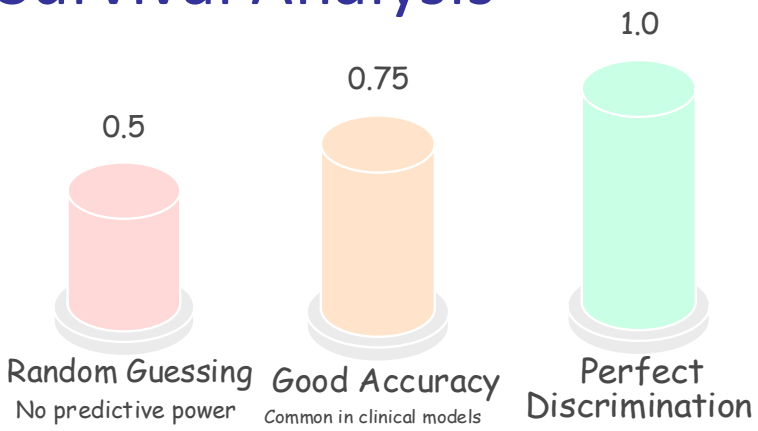
GitHub

[1] Breslow, N. E. (1975). "Analysis of Survival Data under the Proportional Hazards Model". International Statistical Review / Revue Internationale de Statistique. 43 (1): 45–57. doi:10.2307/1402659

Survival Analysis Evaluation

Evaluating Model Performance in Survival Analysis

- **Concordance Index (C-Index)[1]**
  - *Definition:* A global measure of the model's ability to correctly rank pairs of individuals based on their predicted risk.



[1] Antolini, L., Boracchi, P., & Biganzoli, E. (2005). A time-dependent discrimination index for survival data. Statistics in medicine, 24(24), 3927-3944.
[2] Brier, Glenn W. "Verification of forecasts expressed in terms of probability." Monthly Weather Review 78.1 (1950): 1-3.

Evaluating Model Performance in Survival Analysis

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Brain Cancer Analysis

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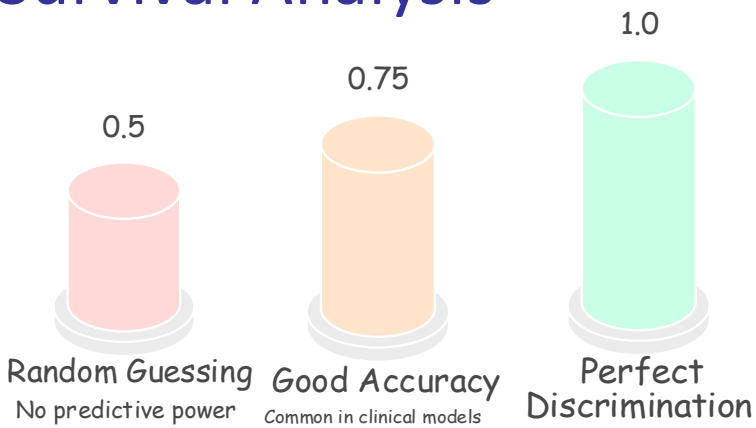
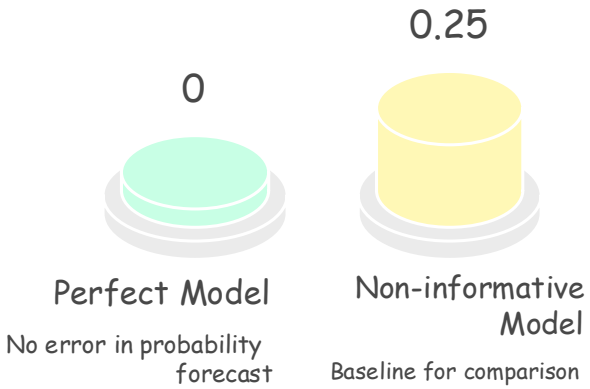


- **Integrated Brier Score (IBS)[2]**
  - While the C-Index measures discrimination (ranking), the IBS measures **calibration** (how close the predicted probability is to the actual outcome).
  - *Definition:* It is the mean squared error of the probability forecast at a specific time point. The "Integrated" version averages this error across the entire study duration.



[1] Antolini, L., Boracchi, P., & Biganzoli, E. (2005). A time-dependent discrimination index for survival data. Statistics in medicine, 24(24), 3927-3944.
[2] Brier, Glenn W. "Verification of forecasts expressed in terms of probability." Monthly Weather Review 78.1 (1950): 1-3.

Additional Use Case

Use Case 2: Publication Bias

- It is a dataset of clinical trials funded by the NHLBI. Our goal is to determine if there is a publication bias: *do trials with statistically significant results reach the journals faster than those with neutral findings?*

Publication Use Case

GitHub

[1] Gordon et al. (2013) Publication of trials funded by the National Heart, Lung, and Blood Institute. New England Journal of Medicine, 369(20):1926–1934.

Use Case 2: Publication Bias

- Covariates
  - *clinend:* whether the trial focused on a clinical endpoint
  - *multi:* Whether the trial involved multiple centers
  - *mech:* The funding mechanism within the National Institutes of Health
  - *sampsize:* Trial sample size
  - *budget:* Budget
  - *impact:* Impact related to the number of citations
  - *Posres:* Whether the trial produced a positive or significant result

	posres	multi	clinend	mech	sampsize	budget	impact	time	status
1	0	0	1	R01	39876	8.016941	44.016	11.203285	1
2	0	0	1	R01	39876	8.016941	23.494	15.178645	1
3	0	0	1	R01	8171	7.612606	8.391	24.410678	1
4	0	0	1	Contract	24335	11.771928	15.402	2.595483	1
5	0	0	1	Contract	33357	76.517537	16.783	8.607803	1

Use Case 2: Publication Bias

Let's follow the same process…

Publication Use Case

GitHub

What have we learnt?

Summary

- When the question is “when” → Survival Analysis
- Censoring is the key difference vs. standard regression
- Survival function and hazard function are related
- Kaplan-Meier estimator gives a non-parametric baseline and group comparisons
- Cox Proportional Hazards model adds covariates and interpretable risk
- Validation matters!! → Chose metrics that respect censoring